Σ^0 Selection for the MicroBooNE Hyperon Analysis

PeLEE ntuples, RHC — selection, reconstruction, and BDT

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Abstract

This note documents the event selection for the $\bar{\nu}_\mu$ charged-current Σ^0 ($\Sigma^0 \to \Lambda\gamma$, $\Lambda \to p\pi^-$) search in MicroBooNE, built on standard PeLEE ntuples in reverse-horn-current (RHC) running. The selection identifies the muon, proton, pion and photon candidates, reconstructs the Λ and Σ^0 invariant masses, and applies a gradient-boosted decision tree (BDT) as the final discriminant, evaluated with two-fold cross-validation. The final selection keeps 502 raw signal events (12.1% of the true Σ^0 in the fiducial volume) at a significance $S/\sqrt{S+B}$ of 0.0537, a 19.5% improvement over the cut-based selection. The surviving background is dominated by resonant and deep-inelastic neutrino interactions with π^0 production.

1 Signal and samples

The signal is the production of a Σ^0 hyperon in $\bar{\nu}_\mu$ CC interactions,

$$\bar{\nu}_\mu + \mathrm{Ar} \;\to\; \mu^+ + \Sigma^0 + X, \qquad \Sigma^0 \to \Lambda\,\gamma, \qquad \Lambda \to p\,\pi^-,$$

giving a final state of a μ^+, a displaced $p\pi^-$ vertex from the Λ decay, and a photon from the Σ^0. Signal events are flagged at truth level by nu_pdg $= -14$, ccnc $= 0$ and a Σ^0 (pdg $= 3212$) in the final-state list, with the true vertex inside the fiducial volume.

Table 1: Samples and POT normalisation. All samples are scaled to the full RHC data exposure. The background uses all five RHC overlay runs.

Sample	Source	Generated POT	Scale factor
Signal (Σ^0)	NuWro hyperon, NoCR	2.1764×10^{23}	0.0053
Other hyperons	NuWro hyperon, NoCR	2.1764×10^{23}	0.0053
Background	GENIE RHC overlay (Run 1+2+4a+4b+4c)	9.9448×10^{21}	0.1159
Dirt	GENIE RHC dirt (Run 1)	4.2127×10^{20}	2.7360
Target data exposure		1.1526×10^{21}	

2 Candidate reconstruction

Particle-flow objects (PFPs) are split into tracks (trk_score > 0.5) and showers. The candidates are chosen as:

- **Muon**: the longest track with trk_score > 0.8, distance to vertex < 20 cm, length > 10 cm, and LLR PID > 0.7.

- **Proton / pion (joint assignment)**: among the remaining tracks a combined PID discriminant is formed,

$$D = \mathrm{LLR}/0.591 + (\chi^2_p - \chi^2_\pi)/124.5,$$

(high D = pion-like, low = proton-like); the proton is the track with $\min D$ and the pion the remaining track with $\max D$. This combination separates true protons from true pions with AUC 0.873, versus 0.838 for χ^2_p alone and 0.852 for the LLR alone (the standardisation scales are taken from truth-matched tracks). The LLR is always defined, so no fallback is needed; the χ^2 term is dropped per track where undefined. This reduces proton contamination of the pion candidate from 35% (original "max-LLR" pion) to 28% (true-pion fraction 31%), and raises the proton-candidate purity to 58%.

- **Photon**: the shower closest to the vertex (shower 1); a second shower (shower 2) is also retained for π^0 rejection.

Λ and Σ^0 invariant masses. The proton momentum is obtained from its range-based kinetic energy (proton hypothesis), the pion momentum from the range momentum (MIP hypothesis), and directions from `trk_dir`. The photon four-vector uses the *per-PFP* shower energy `shr_energy_y_v` (which, unlike the scalar shower-energy branch, is populated) and the unit direction `shr_px/py/pz_v`:

$$m_\Lambda = \sqrt{(E_p + E_\pi)^2 - |\vec{p}_p + \vec{p}_\pi|^2}, \qquad m^0_\Sigma = \sqrt{(E_\Lambda + E_\gamma)^2 - |\vec{p}_\Lambda + \vec{p}_\gamma|^2}.$$

The reconstructed signal Λ peaks at 1.111 GeV (PDG 1.1157 GeV); the Σ^0 mean lands on the PDG value 1.1926 GeV.

3 Selection chain

The cumulative cut chain (and the surviving yields) is given in Table 2. The first eight stages are simple requirements; the ninth is the BDT. The old hard discriminating cuts (proton–muon distance, photon shower size, Λ-mass window) were removed — they are now BDT inputs, so their power is retained and combined optimally.

Table 2: Cut flow. "Raw Σ^0" is the unweighted signal count; efficiency is relative to the true Σ^0 in the fiducial volume; S, B are POT-weighted signal and total background; $S/\sqrt{S+B}$ is the significance.

Cut	Raw Σ^0	Eff. [%]	S	B	Pur. [%]	$S/\sqrt{S+B}$
AllEvents	4148	100.00	21.967	691965.9	0.003	0.0264
FiducialV	3375	81.36	17.874	121860.4	0.015	0.0512
CosmicVeto	2969	71.58	15.724	74052.9	0.021	0.0578
NShowers	2333	56.24	12.355	33188.5	0.037	0.0678
MuonCandidate	2097	50.55	11.106	22927.4	0.048	0.0733
ProtonCandidate	1449	34.93	7.674	18725.0	0.041	0.0561
PionCandidate	682	16.44	3.612	10571.1	0.034	0.0351
NTracksCut	650	15.67	3.442	8261.8	0.042	0.0379
BDTG	**502**	**12.10**	**2.659**	**2452.3**	**0.108**	**0.0537**

The discrete cuts are: fiducial volume (vertex inside the TPC with the dead z region 675.1–775.1 cm removed); cosmic veto topological_score > 0.20; shower multiplicity $0 < n_{\mathrm{shr}} < 4$; muon/proton/pion candidate existence; track multiplicity $n_{\mathrm{trk}} < 5$.

4 BDT discriminant

A gradient-boosted BDT (TMVA, `BoostType=Grad`, $N_{\text{trees}} = 600$, `MaxDepth`=3, `Shrinkage`=0.06, `MinNodeSize`=3%, bagged fraction 0.5) is trained at the NTracksCut preselection on 49 candidate-level variables — track PID/length/score/hits, inter-candidate distances and opening angles, the Λ and Σ^0 masses, shower energy/size, Λ/γ vertex-pointing variables, and π^0 variables (di-photon mass, opening angle, second-shower energy).

Two-fold cross-validation. To avoid applying the BDT to its own training events, two folds are trained — fold A on even event numbers, fold B on odd — and each event is scored by the fold that did *not* train on it. Per-fold ROC integrals are 0.822 and 0.809. The most important variables are the engineered ones: m^0_Σ, proton track score, γ-pointing angle, m_Λ, pion PID, and the soft-photon shower size; the π^0 di-photon mass also contributes.

Operating point. The threshold BDTG > -0.90 is chosen signal-first: it keeps 502 raw signal events (above the cut-based 497) while improving the significance by 19.5% over the cut-based selection ($S/\sqrt{S+B} = 0.0537$ vs 0.0449). The significance is flat over a broad threshold range and reaches ≈ 0.061 at higher purity if more signal can be traded.

5 Background characterisation

Table 3 breaks the surviving background down by neutrino interaction mode. It is dominated by pion-producing resonant (RES) and deep-inelastic (DIS) events: the $\pi^0 \to \gamma\gamma$ supplies a fake Σ^0 photon and the charged hadrons fake the proton/pion tracks. Cosmics are only 3% and are rejected hardest by the BDT.

Table 3: Surviving background by interaction type (final selection), raw MC events and POT-scaled yield.

Mode	Raw	Scaled	% of bkg
Resonant (RES)	12549	1381.8	52.3
Deep inelastic (DIS)	5576	635.3	24.0
Quasi-elastic (QE)	2543	363.8	13.8
Meson exch. (MEC)	1154	177.1	6.7
Cosmic	618	74.6	2.8
Coherent (COH)	71	8.9	0.3
Total	22819	2643.1	100

Candidate purity (background, truth-matched). The muon candidate is a true muon 81% of the time; the proton is a true proton 58%; the pion is true $\pi^\pm$ 31% (another proton 28%); the photon shower is a true photon 29% (a proton 18%). The signal sample is cosmic-free by construction (NoCR), and its non-muon truth matching is unreliable, so only the background purities are quoted.

6 Control distributions

Figure 1 shows the key discriminants at the preselection stage.

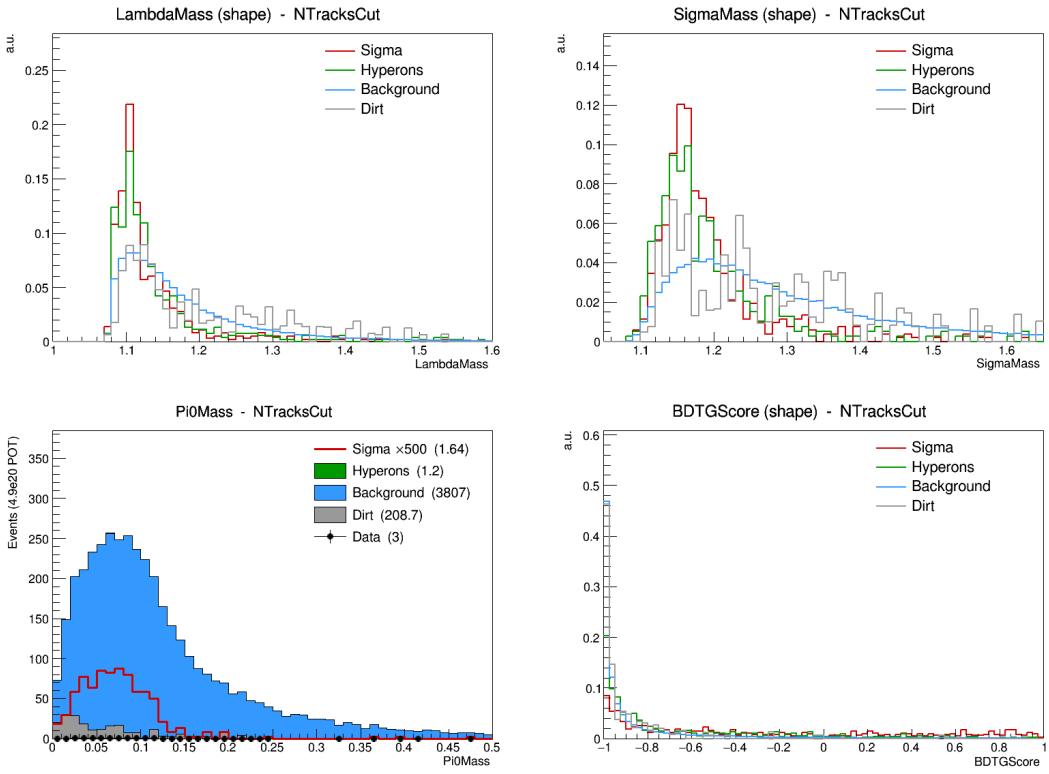


Figure 1: Preselection-stage distributions: reconstructed Λ mass (top left) and Σ^0 mass (top right) shapes; π^0 di-photon mass stack (bottom left); cross-validated BDT score, signal vs background (bottom right).

7 Summary and limitations

The selection combines a physics-driven candidate reconstruction ($\Lambda \to p\pi^-$, $\Sigma^0 \to \Lambda\gamma$ masses) with a cross-validated BDT to reach $S/\sqrt{S+B} = 0.0530$ at 12.2% signal efficiency. The analysis is now *signal-statistics limited*: with 650 signal events at preselection the per-fold ROC plateaus at ≈ 0.81 irrespective of the (much larger) background sample or additional input variables. The largest remaining lever is more signal Monte Carlo. Secondary improvements would come from real shower-energy calibration (to sharpen the Σ^0 mass) and a dedicated π^0 veto using fully reconstructed di-photon events.

Software. Selection: `SigmaSelection.C` (+`SigmaSelConfig.h`, `SigmaFV.h`); BDT training: `TrainBDT.C` (two folds `SigmaBDT_A/`, `SigmaBDT_B/`); plotting: `PlotSigma.C`.